习题 1 使用链式法则计算$\mathrm{d}z/\mathrm{d}t$或$\mathrm{d}\omega/\mathrm{d}t$:

(1) $z = x^2 + y^2 + xy$, $x = \sin t$, $y = e^t$;

(2) $z = \cos(x + 4y)$, $x = 5t^4$, $y = 1/t$;

(3) $z = \sqrt{1 + x^2 + y^2}$, $x = \ln t$, $y = \cos t$;

(4) $z = \arctan(y/x)$, $x = e^t$, $y = 1 - e^{-t}$;

(5) $\omega = xe^{y/z}$, $x = t^2$, $y = 1 - t$, $z = 1 + 2t$;

(6) $\omega = \ln\sqrt{x^2 + y^2 + z^2}$, $x = \sin t$, $y = \cos t$, $z = \tan t$.

习题 2 使用链式法则计算$\partial z/\partial s$或$\partial z/\partial t$:

(1) $z = x^2y^3$, $x = s\cos t$, $y = s\sin t$;

(2) $z = \arcsin(x - y)$, $x = s^2 + t^2$, $y = 1 - 2st$;

(3) $z = \sin\theta\cos\phi$, $\theta = st^2$, $\phi = s^2t$;

(4) $z = e^{x+2y}$, $x = s/t$, $y = t/s$;

(5) $z = e^r\cos\theta$, $r = st$, $\theta = \sqrt{s^2 + t^2}$;

(6) $z = \tan(u/v)$, $u = 2s + 3t$, $v = 3s - 2t$.

习题 3 请画出计算下列函数偏导函数的链式法则示意图, 其中各个位置上的函数均可微:

(1) $u = f(x, y)$, $x = x(r, s, t)$, $y = y(r, s, t)$;

(2) $R = f(x, y, z, t)$, $x = x(u, v, w)$, $y = y(u, v, w)$, $z = z(u, v, w)$, $t = t(u, v, w)$;

(3) $\omega = f(r, s, t)$, $r = r(x, y)$, $s = s(x, y)$, $t = t(x, y)$;

(4) $t = f(u, v, w)$, $u = u(p, q, r, s)$, $v = v(p, q, r, s)$, $w = w(p, q, r, s)$.

习题 4 使用链式法则计算下列多元函数的偏导函数:

(1) $z = x^2y^3$, $x = s\cos t$, $y = s\sin t$;

(2) $z = \arcsin(x - y)$, $x = s^2 + t^2$, $y = 1 - 2st$;

(3) $z = \sin\theta\cos\phi$, $\theta = st^2$, $\phi = s^2t$;

(4) $z = e^{x+2y}$, $x = s/t$, $y = t/s$;

(5) $z = e^r\cos\theta$, $r = st$, $\theta = \sqrt{s^2 + t^2}$;

(6) $z = \tan(u/v)$, $u = 2s + 3t$, $v = 3s - 2t$.

习题 5 使用链式法则计算下列多元函数的偏导函数:

(1) $z = x^4 + x^2y$, $x = s + 2t - u$, $y = stu^2$, $\dfrac{\partial z}{\partial s}$, $\dfrac{\partial z}{\partial t}$, $\dfrac{\partial z}{\partial u}$;

(2) $T = \dfrac{v}{2u+v}$, $u = pq\sqrt{r}$, $v = p\sqrt{q}r$, $\dfrac{\partial T}{\partial p}$, $\dfrac{\partial T}{\partial q}$, $\dfrac{\partial T}{\partial r}$;

(3) $\omega = xy + yz + zx$, $x = r\cos\theta$, $y = r\sin\theta$, $z = r\theta$, $\dfrac{\partial \omega}{\partial r}$, $\dfrac{\partial \omega}{\partial \theta}$;

(4) $P = \sqrt{u^2 + v^2 + \omega^2}$, $u = xe^y$, $v = ye^x$, $\omega = e^{xy}$, $\dfrac{\partial P}{\partial x}$, $\dfrac{\partial P}{\partial y}$.

习题 6 计算下列方程决定的隐函数的导函数$\mathrm{d}y/\mathrm{d}x$:

(1) $y\cos x = x^2 + y^2$;

(2) $\cos(xy) = 1 + \sin y$;

(3) $\arctan(x^2y) = x + xy^2$;

(4) $e^y \sin x = x + xy$.

习题 7 计算下列方程决定的隐函数的偏导函数$\partial z/\partial x$及$\partial z/\partial y$:

(1) $x^2 + 2y^2 + 3z^2 = 1$;

(2) $x^2 - y^2 + z^2 - 2z = 4$;

(3) $e^z = xyz$;

(4) $yz + x\ln y = z^2$.

习题 8 我们称一个二阶连续可微的二元函数$f(x,y)$是**n阶齐次的**, 如果它对所有的t满足如下方程:

$$f(tx, ty) = t^n f(x,y), n \in \mathbb{Z}^+$$

(1) 验证函数$f(x,y) = x^2y + 2xy^2 + 5y^3$是3阶齐次的;

(2) 如果函数$f(x,y)$是n阶齐次的, 请使用链式法则验证如下方程成立:

$$x\frac{\partial f}{\partial x} + y\frac{\partial f}{\partial y} = nf(x,y);$$

(3) 如果函数$f(x,y)$是n阶齐次的, 请验证如下方程成立:

$$x^2\frac{\partial^2 f}{\partial x^2} + 2xy\frac{\partial^2 f}{\partial x \partial y} + y^2\frac{\partial^2 f}{\partial y^2} = n(n-1)f(x,y).$$

(4) 如果函数$f(x,y)$是n阶齐次的, 请验证

$$f_x(tx, ty) = t^{n-1} f_x(x,y).$$